

- Q.26 Write a note on prosthetic cardiac valves.
- Q.27 Explain different ventilation modes.
- Q.28 Write a short note on gas pipe-lines and vacuum system in hospitals.
- Q.29 Explain IABP with diagram.
- Q.30 Write a note on Exercise Stress testing (TMT)
- Q.31 Explain Vascular stents and its types.
- Q.32 Explain syringe pump working with diagram.
- Q.33 Explain ESU with diagram.
- Q.34 Write a short note on nebulizers and humidifiers.
- Q.35 Explain OT table with diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain Telemetry system and its classification with diagram.
- Q.37 Explain Anaesthesia work station with diagram.
- Q.38 Write note on following:
- a) Artificial Heart Valve
 - b) Bag-vent Mask

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183244/033261

Roll No.

4th Semester/ Med. Eltx.
Subject : Biomedical Instrumentation-II/
Biomed. Instrumentation

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which ventilation mode delivers preset tidal volume at preset rate regardless patient effort.
- a) SIMV
 - b) Assist Control
 - c) CPAP
 - d) Pressure support
- Q.2 Which parameter is primarily controlled in pressure controlled ventilation.
- a) Volume
 - b) Time
 - c) Pressure
 - d) Flow
- Q.3 Which sensor is used in ventilator to measure air flow
- a) ECG sensor
 - b) Thermistor
 - c) Flow sensor
 - d) Pulse oximeter

Q.4 Which of the following is Non-Invasive ventilation

- a) CPAP b) SIMV
- c) PCV d) AC

Q.5 The Purpose of defibrillator is

- a) Monitor blood flow b) Detect Arrhythmia
- c) Restart a heart beat d) Measure pulse

Q.6 What is normal range (cm of H₂O) for PEEP

- a) 0-5 b) 5-10
- c) 10-15 d) 20-30

Q.7 Which instrument used for electric shock delivery

- a) Pacemaker b) Stethoscope
- c) Pulse oximeter d) Defibrillator

Q.8 Holder monitoring used for measurement of

- a) Stress ECG b) Continuous ECG
- c) Lung capacity d) Blood pressure

Q.9 A catheter is used for following

- a) Angioplasty b) Stress test
- c) Echo-Cardiography d) None

Q.10 Which part of pacemaker is implanted inside heart

- a) Battery b) Pulse Generator
- c) Lead or Electrode d) Microprocessor

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 Expand NICU.

Q.12 Telemetry.

Q.13 Prosthetic

Q.14 Expand IABP

Q.15 Respiration Rate.

Q.16 Expand FiO₂.

Q.17 Minute Volume

Q.18 Fibrillation.

Q.19 Cardiac output.

Q.20 Pacemaker.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write a note on classification of orthotics & prosthetic.

Q.22 Explain tele-medicine and its application with diagram.

Q.23 Explain multichannel telemetry system with diagram.

Q.24 Explain scavenging system in Operation theatre.

Q.25 Explain need of artificial ventilation and types of mechanical ventilation.